TING-WEI HSU

(+1) 404-734-4484 twhsu@gatech.edu LinkedIn Google Scholar

RESEARCH INTERESTS

RF Sensing, mmWave Systems, Wireless Digital Twin, Datacenter Networks, LLM Serving, Deep Learning for Communications

EDUCATION

Georgia Institute of Technology Atlanta, USA
Ph.D. in Electrical and Computer Engineering *Aug. 2024 – Present (expected May 2029)*
Advisor: Prof. Karthikeyan Sundaresan GPA: 4.0/4.0

National Taiwan University Taipei, Taiwan
Master of Science in Communication Engineering *Sept. 2021 – Feb. 2024*
Advisor: Prof. Hung-Yun Hsieh GPA: 4.3/4.3
Thesis Title: Multi-User Pose Estimation Based on Spatial and Temporal Features from WiFi CSI

Keio University Tokyo, Japan
Exchange student *Sept. 2023 – Feb. 2024*

National Taiwan University Taipei, Taiwan
Bachelor of Science *Sept. 2016 – Jan. 2021*
Major: Mechanical Engineering GPA: 3.89/4.3
Minor: Computer Science and Information Engineering

PUBLICATIONS

* Equal contribution.

[1] **T. -W. Hsu**, K. Sundaresan, "VLM-driven Digital Twin," Under review.
[2] **T. -W. Hsu***, Y. -T. Lin*, K. Sundaresan, "Vision-to-RF Digital Twin for mmWave networks," Under review. [website]
[3] Y. -T. Lin*, **T. -W. Hsu***, K. Sundaresan, "Traffic Engineering for mmWave Networking in Industrial AI," Under review. [website]
[4] Y. -T. Lin*, **T. -W. Hsu***, K. Sundaresan, "Distributed Collaborative Positioning through ISAC for GPS-Denied Robotic Swarms," ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc), 2026 (to appear). [website]
[5] H. -C. Lin*, **T. -W. Hsu***, C. -E. Ho, A. Saeed, "ARK: Avoiding Routing Collisions for KV Cache Transfer in Disaggregated LLM Inference," ACM SIGCOMM Workshop on Networks for AI Computing (NAIC), Denver, CO, USA, 2026 (to appear).
[6] **T. -W. Hsu**, H. -Y. Hsieh, "On Using Spatial and Temporal Features for Robust Multiuser Pose Estimation Based on WiFi CSI," IEEE Internet of Things Journal, vol. 12, no. 15, pp. 29897–29912, Aug. 2025. [paper]
[7] **T. -W. Hsu**, H. -Y. Hsieh, "Robust multi-user pose estimation based on spatial and temporal features from WiFi CSI," 2024 IEEE International Conference on Communications (ICC), Denver, CO, USA, 2024. [paper]
[8] **T. -W. Hsu** and J. -S. Liu, "Design of smooth path based on the conversion between η^3 spline and Bézier curve," 2020 American Control Conference (ACC), Denver, CO, USA, 2020. [paper]
[9] **T. -W. Hsu** and J. -S. Liu, "Convex hull determination method for generating Bézier curve followed by Trailer-tractor System," Automation 2020, Hualien, Taiwan, November 2020. [paper]

RESEARCH EXPERIENCE

Mobile Advanced Research, GaTech *Aug. 2024 – Present*
Graduate Research Assistant Advisor: Prof. Karthikeyan Sundaresan

· Built vision-to-RF mmWave digital twins [1, 2] predicting beamformed link profiles from RGB-D vision, cutting RF site-survey overhead up to 61× and reaching <4 dB RSSI error on unseen sites.
· Developed a digital-twin-driven predictive traffic-engineering framework [3] assuring industrial real-time-AI networking ∼100% of the time (vs. ≤35% baseline), with 1.3–1.5× throughput and ∼5× lower disruption.
· Developed DiCoP [4], an infrastructure-free ISAC collaborative-positioning system for GPS-denied swarms, achieving sub-2 m accuracy for ∼89% of nodes and scaling to 100 UAVs.

Mobile Networks and Wireless Communications Lab, NTU *Sept. 2021 – Aug. 2024*
Graduate Research Assistant Advisor: Prof. Hung-Yun Hsieh

· Designed a robust multi-user pose-estimation framework from WiFi CSI, improving cross-domain accuracy by ∼12% (mPCK) over prior methods while running in real time (∼19 FPS) [6, 7].
· Proposed three novel approaches to capture spatial and temporal features from WiFi CSI.
· Set up CSI simulation and real experiment platform for experiment and verification.

Automation Lab, Academia Sinica *Sept. 2019 – Dec. 2020*
Part-Time Research Assistant Advisor: Prof. Jing-Sin Liu

· Derived η^3-spline / Bézier path transformations and curvature-constrained trajectory design for trailer-tractor parking [8, 9].

Bio-Inspired Robotic Lab, NTU *Sept. 2018 – Aug. 2020*
Undergraduate Research Assistant Advisor: Prof. Pei-Chun Lin

· Simulated and analyzed jumping-robot gait patterns in MATLAB.
· Designed and manufactured the robot leg mechanism and tuned its brushless-motor PID controller.

TEACHING EXPERIENCE

Teaching Assistant, Computer Communications, GaTech *Fall 2024, Spring 2025*
Teaching Assistant, Computer Programming (×2) & Probability and Statistics, NTU *2021 – 2022*

HONORS AND COMPETITION EXPERIENCE

JASSO Scholarship *Sept. 2023*
· Merit-based scholarship for foreign students, awarded by the Japanese government.

HITCON World CTF Competition *Jan. 2023*
· Top 40%; responsible for the cryptography section.

SELECTED PROJECTS

ARK: Collision-Free KV Cache Transfer for LLM Inference, CS 8803 Datacenter Network Systems, GaTech *Spring 2026*
· Built a packet-level NS-3 simulation of KV-cache transfer for disaggregated LLM inference and designed a distributed source-port path reservation that avoids ECMP hash collisions, cutting P95 flow-completion time up to 34% [5].

Real-Time Channel Depth Detection Project, Introduction to IoT, NTUGICE *Fall 2021*
· Built a real-time channel-depth detection system with deep learning, in collaboration with MiTAC.

SKILLS

Programming: Python, C/C++, MATLAB, LaTeX
ML / DL: PyTorch, NumPy/SciPy, Vision-Language Models, Neural Radiance Fields (NeRF)
Wireless / RF: mmWave testbeds, WiFi CSI, USRP / GNU Radio, ns-3, ray tracing
Languages: Chinese (native), English, Japanese